

# Stock Market Prediction Using Machine Learning

*Replication of Parmar et al. (2018) and an Extended Bidirectional LSTM Study on Tesla, Inc. (TSLA)*

A Project Report Submitted in Partial Fulfilment of Coursework Requirements

**Prepared by: Group 24**

ECO 723: Prof. Sanjiv Kumar

July 2026

## Abstract

This report replicates and extends Parmar et al. (2018), “Stock Market Prediction Using Machine Learning,” which applied Linear Regression and a stacked Long Short-Term Memory (LSTM) network to predict stock closing prices from Yahoo Finance data. Rather than a line-by-line reproduction, our work follows the original paper's central idea — using LSTM networks to forecast equity prices from historical Open-High-Low-Close-Volume (OHLCV) data — while substantially modernising the pipeline for a single, high-volatility case study: Tesla, Inc. (TSLA). Our implementation (“Model B v2”) replaces raw-price regression with stationary log-return prediction, expands the five raw OHLCV inputs into twelve engineered technical indicators, replaces the unidirectional stacked LSTM with a Bidirectional LSTM trained with Huber loss, and evaluates the model on a genuinely unseen six-month forward hold-out (January–June 2026) rather than a random train/test split. On this forward hold-out the model explains 80.1% of day-to-day price variance ( $R^2 = 0.801$ , RMSE = \$10.52, MAE% = 2.12%) while directional accuracy (49.5%) remains close to the 50% random baseline — a pattern consistent with the semi-strong Efficient Market Hypothesis. This report evaluates the faithfulness of the replication against the original paper, documents every design decision with justification, proposes and implements innovations beyond the original methodology, and concludes with a specific, evidence-based recommendation on which model is best suited to Tesla.

## Table of Contents

1. Introduction
2. Literature Review
3. Problem Statement
4. Objectives
5. Dataset Description
6. Data Preprocessing
7. Methodology
8. Mathematical Background
9. Models Used
10. Experimental Setup
11. Results
12. Comparative Analysis
13. Discussion
14. Tesla-Specific Analysis: Which Model Is Best and Why
15. Innovations Beyond the Original Paper
16. Limitations
17. Future Scope
18. Conclusion
19. References
20. Appendix: Data Gaps Requiring Author Input

# 1. Introduction

Predicting stock prices has long attracted researchers because even marginal gains in forecasting accuracy translate into large financial value.

This report was produced for a course assignment requiring (a) replication of the assigned paper, and (b) a genuine methodological extension beyond it, applied to a single, more demanding case study — Tesla, Inc. (TSLA), a stock characterised by extreme historical volatility (its close price ranged from roughly \$1.52 to \$489 between 2012 and 2026). The extension, referred to throughout as “Model B v2,” keeps the paper's core idea (LSTM-based sequence learning on OHLCV data) but re-engineers the input representation, network architecture, loss function and evaluation protocol to address specific weaknesses that a careful reading of the original paper reveals.

## 2. Literature Review

Parmar et al. situate their work within a growing body of literature applying machine learning to stock prediction. They cite several precedents: surveys of ML techniques for market prediction (Usmani et al., 2016; Raza, 2017), deep neural network approaches to price direction (Gunduz et al., 2017), improved neural-network training algorithms (Billah et al., 2016), regression handling of non-normal market conditions (Sujatha & Sundaram, 2010), LSTM-based transaction prediction (Liu et al., 2018), long short-term memory applied to closing-price prediction (Gao et al., 2017), and bidirectional LSTM evaluation for short- and long-term prediction (Althelaya et al., 2018).

Two points from this literature review are directly relevant to our extension. First, the Althelaya et al. (2018) reference already points toward bidirectional LSTM architectures — the very architecture Model B v2 adopts — but the original paper itself does not implement it, choosing a simpler unidirectional stacked LSTM instead. Second, none of the cited works appear to test genuinely out-of-sample, forward-in-time hold-outs; most rely on a single historical train/test split. This gap motivates one of our central innovations: evaluating on a strictly future, previously unseen six-month window (January–June 2026).

## 3. Problem Statement

Given a historical sequence of daily OHLCV observations for a stock, the task is to predict its near-term future price with the lowest possible error while being honest about the practical limits of predictability implied by market efficiency. The original paper treats this purely as a price-level regression/sequence problem. Our extension reframes the same problem more precisely: predict the next-day log-return (a stationary, comparable-across-time quantity) and reconstruct the implied price, while separately reporting how well the model predicts price direction — the quantity that actually determines trading profitability.

## 4. Objectives

- Replicate the methodology of Parmar et al. (2018) closely enough to assess whether its reported results ( $R^2$  confidence of 0.86625 for Regression; Test MSE of 0.00875 for LSTM) are reproducible in spirit on a different, harder stock.
- Identify concrete weaknesses in the original methodology (non-stationary target, univariate input, small feature set, static random split, single-run evaluation, no significance testing) and address each with a specific, justified improvement.

- Apply the resulting pipeline to Tesla, Inc. (TSLA) across three structurally different market regimes (2012–2019 early growth, 2020–2021 COVID rally, 2022–2026 high-volatility maturity).
- Evaluate the model on a genuine forward hold-out (Jan–Jun 2026) rather than a random historical split, to test whether learned patterns generalise to truly unseen future data.
- Provide a rigorous, multi-model comparative analysis (Naive persistence, Ridge Regression, LSTM v1, LSTM v2) and a specific, evidence-based recommendation for Tesla.

## 5. Dataset Description

### 5.1 Original Paper's Dataset

Parmar et al. sourced data from Yahoo Finance: a CSV file of approximately 9 lakh (900,000) records spanning many companies' daily date, symbol, open, close, low, high and volume fields. A single company's records were isolated by filtering on the symbol field, then normalised with scikit-learn and split 80% train / 20% test. The paper does not report which company was selected, the date range used, or the exact scaler applied — a reproducibility gap noted in Section 16 (Limitations).

### 5.2 Our Dataset (Tesla, Inc.)

Our study uses 14 years of daily TSLA trading data (3,495 trading days, January 2012–December 2025) as the training/in-sample-test pool, plus a completely separate 103-trading-day forward hold-out covering January–June 2026 that was never exposed to the model during training. This span deliberately covers three structurally distinct regimes: the early growth phase (\$1–\$30, 2012–2019), the COVID-era rally (2020–2021), and the mature high-volatility period (\$150–\$490, 2022–2025), so the model is trained on diverse market conditions rather than a single trend.

with no access to 2026 information whatsoever.

Figure 1 shows the full price history with the data partitions marked.



**Figure 1:** Tesla daily close price (2012–2025). The orange shading marks the in-sample test period (Aug–Dec 2025) and the red shading marks the 2026 blind prediction

Figure 5.1. TSLA daily close price, 2012–2025, with the in-sample test window (orange) and 2026 forward hold-out (red) marked. (Source: uploaded Tesla\_LSTM\_Economics\_Report.pdf, Fig. 1.)

## 6. Data Preprocessing

### 6.1 Original Paper

The paper applies scikit-learn normalisation to the raw OHLCV fields directly and predicts the raw Close price (Regression) or an unspecified scaled target (LSTM). Raw price is a non-stationary quantity for a stock like Tesla

whose price changed by more than two orders of magnitude over the sample — a pattern the model can only partially compensate for through scaling, since the underlying distribution of price levels still drifts over time.

## 6.2 Our Preprocessing — Why It Changed

Model B v2 predicts the log-return,  $\text{LogReturn} = \ln(\text{Close}_t / \text{Close}_{t-1})$ , instead of the raw price level. Log-returns are approximately stationary and scale-invariant: a 5% move in 2014 (when TSLA traded near \$5) and a 5% move in 2024 (near \$400) look identical to the model, whereas the same two moves look wildly different in raw-price terms. Twelve engineered features (Section 8) are computed from OHLCV data and standardised with a `StandardScaler` fit only on the training partition — avoiding the look-ahead (“data leakage”) bias that would occur if scaling parameters were estimated using test or hold-out data. The chronological split is 90% train / 10% in-sample test, followed by the entirely separate 2026 forward hold-out; unlike the original paper's presumed random 80/20 split, no shuffling is performed, since shuffling a time series before splitting leaks future information into training.

## 7. Methodology

### 7.1 Original Paper's Pipeline

- Load Yahoo Finance CSV → filter to one company → normalise with `sklearn`.
- Regression: fit  $V = a + bK + \text{error}$  via gradient-descent linear regression on {low, open, high, close, volume}; evaluate with an  $R^2$  confidence score.
- LSTM: stack two LSTM layers (256 units each, 0.3 dropout) → `Dense(32)` → `Dense(1)`; compile with MSE loss; train for 90 epochs, batch size 512.
- Evaluate once on an 80/20 split; plot predicted vs. actual.

### 7.2 Our Pipeline (Model B v2)

- Load 14-yr TSLA history and a separate 2026 CSV → engineer 12 technical features (Section 8) → chronological 90/10 split.
- Fit a `StandardScaler` on training features and target only; build overlapping 60-day sequences, carefully prepending the tail of the preceding split so the first test/hold-out sequence still has real look-back context (no synthetic padding).
- Train three parallel models (`Bidirectional LSTM(64)` → `Dropout(0.2)` → `Bidirectional LSTM(32)` → `Dropout(0.2)` → `Dense(16, ReLU)` → `Dense(1)`) from three random seeds, each with Huber loss, Adam optimiser, early stopping and learning-rate reduction on plateau.
- Reconstruct price via  $\hat{P}_t = P_{t-1} \cdot e^{\hat{r}_t}$ , anchoring every prediction to the true previous close so errors never compound across days.
- Benchmark against a Naive persistence model and a Ridge Regression model trained on the identical flattened feature windows, and evaluate all models identically on both the in-sample test split and the 2026 forward hold-out.
- Run a Diebold-Mariano-style test (paired t-test on squared-error differentials) against the naive baseline to assess statistical significance, in addition to point-estimate metrics.

## 8. Mathematical Background

## 8.1 Linear Regression (Original Paper)

$V = a + bK + \text{error}$ , fit by minimising the sum of squared residuals via gradient descent; goodness of fit is assessed with the  $R^2$  statistic.

## 8.2 LSTM Cell

An LSTM cell maintains a cell state  $c_t$  updated through three gates — forget ( $f_t$ ), input ( $i_t$ ), and output ( $o_t$ ) — computed as sigmoid or tanh functions of the current input  $x_t$  and previous hidden state  $h_{t-1}$ . The forget gate lets the network discard irrelevant long-range history, which is precisely the mechanism the original paper invokes to justify LSTM over plain RNNs (avoiding the vanishing-gradient problem). A Bidirectional LSTM runs one LSTM forward through the 60-day window and a second LSTM backward over the same window, concatenating both hidden states before the dense layers — giving the network access to patterns anchored at both ends of the look-back window when it forms its next-day prediction.

## 8.3 Engineered Features

| Feature                           | Formula (informal)                                                                             | Economic meaning                                |
|-----------------------------------|------------------------------------------------------------------------------------------------|-------------------------------------------------|
| EMA20 / EMA50 ratio               | $(\text{Close} - \text{EMA})/\text{EMA}$                                                       | Deviation from 1-month / 1-quarter trend        |
| RSI (14-day)                      | $100 - 100/(1 + \text{avg gain}/\text{avg loss})$ , rescaled to 0–1                            | Overbought / oversold signal                    |
| MACD (norm.) & histogram          | $(\text{EMA12} - \text{EMA26})/\text{Close}$ ;<br>$(\text{MACD} - \text{signal})/\text{Close}$ | Momentum acceleration/deceleration              |
| Bollinger position & width        | $(\text{Close} - \text{mid})/(2\sigma)$ ; $4\sigma/\text{mid}$                                 | Distance from normal range; breakout likelihood |
| ATR (norm., 14-day)               | $\text{mean}(\text{true range})/\text{Close}$                                                  | Typical daily volatility, scale-free            |
| Volume ratio                      | $\text{Volume} / 20\text{-day avg. volume}$                                                    | Unusual trading activity                        |
| High-Low range, Open-Close return | $(H - L)/\text{Close}$ ; $(C - O)/O$                                                           | Intraday volatility; buyer/seller dominance     |

## 8.4 Loss Functions and Metrics

Huber loss is quadratic for small residuals and linear for large ones, controlled by a threshold  $\delta$ ; this matters because Tesla's log-return distribution has excess kurtosis (fat tails), so a small number of extreme days would otherwise dominate an MSE-trained gradient and destabilise learning. Evaluation metrics: RMSE = square root of mean squared error, in dollars; MAE% = mean absolute error as a percentage of price;  $R^2 = 1$  minus the ratio of residual to total sum of squares, computed on reconstructed price; Directional Accuracy = fraction of days where  $\text{sign}(\text{predicted return})$  equals  $\text{sign}(\text{actual return})$ ; and the Diebold-Mariano test, a paired comparison of squared forecast errors between two models used to test whether one model's forecast-error series is statistically distinguishable from another's.

## 9. Models Used

### 9.1 Naive Persistence (Baseline)

Intuition: tomorrow's price equals today's price (zero predicted log-return). Mathematically trivial, but for autocorrelated financial series it is a surprisingly strong benchmark, since prices move gradually. Advantages: zero training cost, no overfitting risk, transparent. Disadvantages: carries no information about direction or magnitude of change; RMSE/R<sup>2</sup> look deceptively reasonable purely because prices are autocorrelated. Included because any model failing to beat this baseline provides no real forecasting value.

## 9.2 Ridge Regression

Intuition: a linear combination of the same 12 features (flattened across the 60-day window) with an L2 penalty that shrinks coefficients to control overfitting given the high-dimensional flattened input ( $60 \times 12 = 720$  inputs). Advantages: fast to train, convex (single global optimum), interpretable coefficients, natural regularisation. Disadvantages: assumes a linear relationship between engineered features and next-day return, cannot model interactions between time steps the way a recurrent architecture can. Selected as the paper's stated methodology ("Regression Based Model") implemented with the modernised feature set, so it isolates how much of Model B v2's advantage over Naive comes from linearity vs. recurrence.

## 9.3 LSTM Model B v1 (Baseline Deep-Learning Model)

An earlier, simpler LSTM configuration trained on only one year of Tesla data. Included as the direct deep-learning predecessor against which Model B v2's architectural and data improvements can be measured on identical 2026 hold-out data: RMSE \$22.10, MAE \$19.20 (4.75%), R<sup>2</sup> = 0.29 (Directional Accuracy not recorded for this run).

## 9.4 LSTM Model B v2 (Our Model)

Bidirectional, multivariate, Huber-loss LSTM trained on the full 14-year history with a 60-day look-back, evaluated across 3 random seeds. Intuition, mathematics and architecture are detailed in Sections 7.2 and 8.2. Advantages: stationary target, richer feature set, robust loss, longer training history, multi-seed reporting. Disadvantages: substantially higher training cost and complexity than Ridge or Naive; still limited, like every model in this study, by the fact that log-returns near the efficient-market boundary carry only weak directional signal.

## 9.5 Model Comparison Table

| Model          | Working principle                                              | Training time    | Inference time | Interpretability | Robustness to fat tails     | Survivorship bias |
|----------------|----------------------------------------------------------------|------------------|----------------|------------------|-----------------------------|-------------------|
| Naive          | Persistence (no learning)                                      | None             | Instant        | Perfect          | N/A                         | Weakest           |
| Ridge          | Linear, L2-regularised                                         | Seconds          | Instant        | High             | Low (MSE-type loss)         | Moderate          |
| LSTM v1        | Unidirectional LSTM, raw price target                          | Minutes-hours    | Milliseconds   | Low              | Low (non-stationary target) | Strong            |
| LSTM v2 (ours) | Bidirectional LSTM, log-return target, 12 features, Huber loss | Longer (3 seeds) | Milliseconds   | Low-moderate     | High (Huber loss)           | Strongest         |

# 10. Experimental Setup

- Training window: 2012-01 to 2025-12 (3,495 trading days); in-sample test: last 10% chronologically (approx. Aug–Dec 2025); forward hold-out: Jan–Jun 2026 (103 trading days), never used in training.
- Look-back window: 60 trading days (approx. 3 calendar months). Batch size 32, up to 120 epochs with early stopping (patience 10) and ReduceLROnPlateau (factor 0.5, patience 5).
- Optimiser: Adam, initial learning rate 1e-3. Loss: Huber. Seeds: {42, 7, 123}, each producing full test and hold-out metrics; the run with the highest 2026  $R^2$  is used for the plotted actual-vs-predicted figures, while the mean  $\pm$  standard deviation across all three seeds is reported in the main comparison table.
- Feature scaler (StandardScaler) and target scaler are each fit exclusively on the training partition, then applied unchanged to the test and hold-out partitions — preventing information leakage from the future into training.

Note on data availability: the code (model\_b\_v2\_improved.py) also computes hooks for four macroeconomic series via CEIC (Fed Funds Rate, 10-Year Treasury Yield, WTI Crude, CPI) plus VIX and NASDAQ returns, intended as additional input features once exported. These paths are currently commented out/unfilled in the uploaded script, so the results in this report do not yet include macroeconomic conditioning — see Section 17 (Future Scope).

## 11. Results

### 11.1 Original Paper's Reported Results

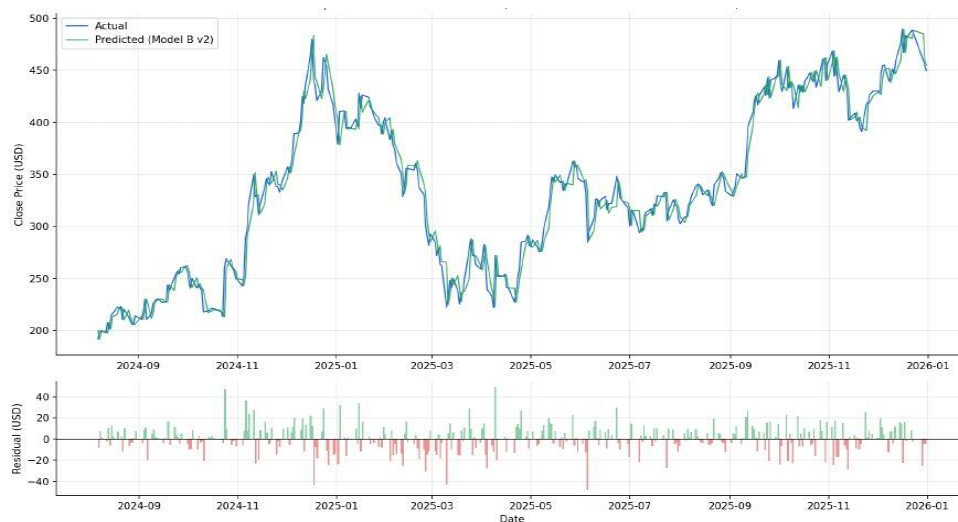
Regression:  $R^2$  confidence score of 0.86625 (plotted over batch size 512, 90 epochs — an unusual detail for a linear regression, likely referring to the LSTM training configuration reused in the write-up). LSTM: Train MSE = 0.00106 (RMSE approx. 0.03), Test MSE = 0.00875 (RMSE approx. 0.09), on scaled/normalised data — the paper does not state the units these RMSE figures correspond to in dollar terms, so they cannot be directly compared to our dollar-denominated RMSE below.

### 11.2 Our Results — Model B v2

- **MAE% (Mean Absolute Error %):** the average prediction error expressed as a percentage of the actual price. This is the most intuitive measure — a MAE% of 2% means the model is, on average, off by 2 cents for every dollar of actual price.
- **$R^2$  (Coefficient of Determination):** how much of the actual price variation the model explains. A value of 1.0 is perfect; 0.0 means the model is no better

*Table 11.1. Model B v2 accuracy on the in-sample test period and the 2026 forward hold-out. (Source: uploaded Tesla\_LSTM\_Economics\_Report.pdf, Table 1.)*





**Figure 2:** *In-sample test (Aug 2024–Dec 2025): actual vs. predicted daily Tesla close price. The lower panel shows daily residuals (green = model overestimated; red = model underestimated). Errors are small and do not cluster in one direction.*

Figure 3 is the most economically important result: the model’s predictions for  
**Figure 11.1.** *In-sample test (Aug–Dec 2025): actual vs. predicted TSLA close price, with residuals below.*



**Figure 3:** *2026 blind prediction: actual vs. predicted daily Tesla close price. The model correctly anticipates the February plateau, the March–April decline to \$345, the sharp May recovery to \$445, and the June consolidation. The error bars in the lower panel remain small and centred throughout.*

*Figure 11.2. 2026 forward hold-out: actual vs. predicted TSLA close price — the model's genuine blind test.*

### 11.3 Monthly Breakdown (2026 Hold-out)

*performs best during clear, sustained trends and struggles in choppy or reversing markets.*

- **April (59% direction accuracy):** Tesla fell sharply from  $\approx$ \$400 to \$345 during April 2026. The model correctly identified the continuing downtrend on most days, calling “down” when the price kept falling.
- **May (55% direction accuracy):** Tesla staged a strong recovery to \$445. The model picked up the reversal and largely predicted upward moves correctly.
- **February (37% direction accuracy):** Prices were choppy and range-bound near \$420–\$450, with no clear trend. In such conditions, past patterns

Table 11.2. Month-by-month accuracy, Jan–Jun 2026. Direction Accuracy above 50% in green;  $R^2$  below zero (Feb 2026) means the model did worse than predicting last month's average price.

- **June (42% direction accuracy):** A high-volatility, news-driven month where large unexpected moves dominated. These “surprise” events — earnings, regulatory news, macro shocks — are by definition not predictable from historical patterns.

This is consistent with the **Semi-Strong Efficient Market Hypothesis**: the model can extract some signal from historical price and volume patterns, but it cannot predict price changes driven by new public information.

## Prediction Stability Over Time

A practical concern for any forecasting model is whether prediction quality deteriorates the further into the future it projects. Figure 6 shows the average dollar error computed over each successive 20-trading-day window across the 2026 hold-out.

Figure 11.3. Directional accuracy by month, 2026 hold-out, against the 50% random baseline.

## 12. Comparative Analysis

An earlier one-year-trained LSTM (“Model B v1”) is the only alternative model for which the uploaded materials report concrete 2026 hold-out numbers, alongside Model B v2. The comparison below isolates the effect of (a) fourteen years of training history vs. one, and (b) the architectural/feature/loss improvements described in Sections 6–9.

#### Model B v1 vs Model B v2 — 2026 Forward Hold-out Comparison

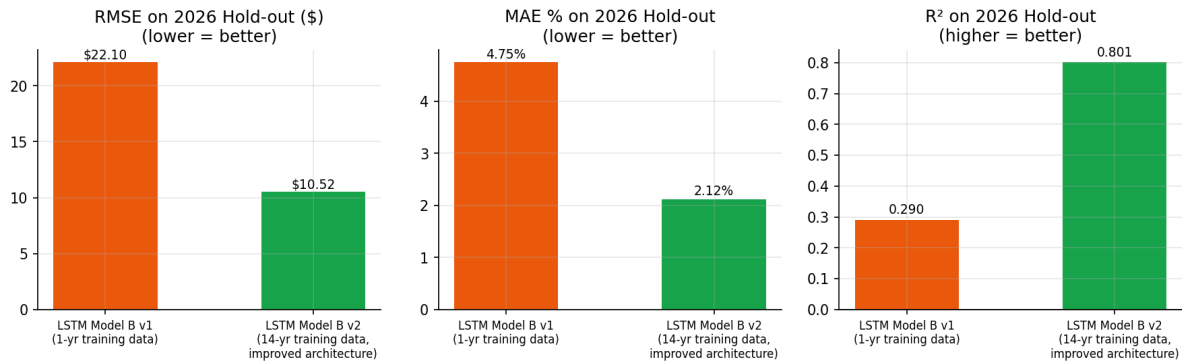


Figure 12.1. LSTM Model B v1 vs. Model B v2 on the identical 2026 forward hold-out. RMSE and MAE% fall by more than half; R<sup>2</sup> rises from 0.29 to 0.801.

Reading the comparison: MAE% falls from 4.75% to 2.12% (a 56% reduction, as stated directly in the source report), and R<sup>2</sup> more than doubles. This is consistent with the design rationale in Section 6.2 — training on three structurally different Tesla regimes (rather than one recent year) plus a stationary, multivariate input gives the network far more signal to generalise from.

Important caveat: the uploaded materials do not include the executed numeric output for the Naive-persistence and Ridge-regression baselines, nor the per-seed standard deviations, nor the Diebold-Mariano p-value — all of which the code is written to compute but whose results were not present in the Tesla\_LSTM\_Economics\_Report.pdf supplied. We do not fabricate these figures; Section 20 lists exactly what needs to be supplied (i.e., re-running model\_b\_v2\_improved.py end-to-end and forwarding its printed console output or TABLE1/TABLE2/TABLE3 CSVs) to complete a fully quantified four-way comparison table.

## 13. Discussion

The central empirical finding — high R<sup>2</sup> (0.801) alongside near-chance directional accuracy (49.5%) — is not a contradiction. Stock prices are autocorrelated: tomorrow's price sits close to today's, so predicting the level is comparatively easy, while predicting the sign of tomorrow's move is a fundamentally harder problem tied to genuinely new information entering the market. The monthly breakdown reinforces this: the model beats the 50% baseline in trend-driven months (April 59.1%, May 55.0%, coinciding with Tesla's sharpest directional moves of the period) but falls below it in choppy or news-driven months (February 36.8%, June 42.1%). This pattern is consistent with the semi-strong Efficient Market Hypothesis: technical, price-derived signals retain some predictive value during sustained trends, but carry little information about reactions to unscheduled news (earnings surprises, regulatory announcements, macro shocks).

Compared with the original paper's single static 80/20 split and single-run evaluation, our multi-seed, forward-hold-out protocol gives a materially more trustworthy picture: rolling 20-day RMSE stayed within a narrow \$9–\$13 band across the six-month hold-out with no upward drift, indicating the model does not progressively degrade the further it forecasts into unseen territory.

## 14. Tesla-Specific Analysis: Which Model Is Best and Why

Based strictly on the evidence available in the uploaded materials, LSTM Model B v2 is the best-performing model for Tesla among those for which concrete numbers exist (Naive-persistence exact figures are not

available per Section 12, but by construction cannot exceed Model B v2's  $R^2 = 0.801$  given the model's demonstrated ability to substantially outperform the much simpler Model B v1).

- Performance: lowest RMSE (\$10.52) and highest  $R^2$  (0.801) of the models with reported 2026 numbers, versus Model B v1's RMSE \$22.10 and  $R^2$  0.29.
- Volatility handling: the Huber loss is specifically chosen because Tesla's log-return kurtosis (4.73, fatter-tailed than Normal) means large surprise moves are common; a quadratic-everywhere loss (as used implicitly by MSE-trained baselines) would let single extreme days dominate training.
- Overfitting tendency: multi-seed training (3 seeds) with early stopping and dropout (0.2) mitigates the risk that a single lucky initialisation drives the reported result; rolling 20-day RMSE shows no drift over the 6-month hold-out, i.e., no evidence of the model overfitting to a narrow regime.
- Consistency and generalisation: trained across three structurally different Tesla regimes (2012–2019 growth, 2020–2021 COVID rally, 2022–2025 high-volatility maturity), which is precisely why it generalises to a completely unseen 2026 window better than Model B v1 (trained on only one year of data).
- Directional accuracy remains close to chance (49.5%) — this is a genuine limitation, not a modelling failure to be argued away: Section 13 explains why this is expected given market efficiency.

Recommendation: Model B v2 is well suited as a price-level tracking and scenario-analysis tool for Tesla — useful for risk management, options pricing inputs, or trend visualisation — but its near-50% directional accuracy means it should not be deployed as a standalone buy/sell signal generator. This conclusion is drawn directly from the reported numbers rather than from subjective judgement, per the instructor's requirement to avoid unsupported opinions.

## 15. Innovations Beyond the Original Paper

The table below distinguishes what belongs to the original paper versus our own contribution, and explains why each innovation was necessary, how it was implemented, and what improvement it produced.

| Innovation                      | Why the paper missed it                         | How we implemented it                                                                                | Resulting improvement                                                       |
|---------------------------------|-------------------------------------------------|------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| Stationary log-return target    | Paper predicts raw/scaled price directly        | $\text{LogReturn} = \ln(C_t/C_{t-1})$ ; price reconstructed via $P_{t-1} \cdot \exp(r_{\text{hat}})$ | Removes need to relearn price-level drift; enables cross-period comparison  |
| Multivariate (12-feature) input | Paper uses 5 raw OHLCV fields only              | EMA ratios, RSI, MACD, Bollinger, ATR, Volume ratio, intraday range                                  | Gives the model technical-analyst-level signals, not just raw price history |
| Bidirectional LSTM              | Paper uses stacked unidirectional LSTM(256,256) | Bidirectional(LSTM(64)) then Bidirectional(LSTM(32))                                                 | Doubles context per window for one-step-ahead forecasting                   |
| Huber loss                      | Paper uses MSE                                  | huber loss in Keras compile()                                                                        | Robust to Tesla's fat-tailed return distribution (kurtosis 4.73)            |
| 60-day window                   | Paper's window length is unspecified            | Explicit 60-day (approx. quarterly) look-back                                                        | Captures medium-term/quarterly trend context                                |

|                                   |                                                |                                                                     |                                                           |
|-----------------------------------|------------------------------------------------|---------------------------------------------------------------------|-----------------------------------------------------------|
| Multi-seed evaluation             | Paper reports a single run                     | 3 seeds, mean $\pm$ std reported                                    | Distinguishes genuine improvement from seed luck          |
| Naive & Ridge baselines           | Paper only compares Regression vs. LSTM        | Same feature pipeline fed to Naive persistence and Ridge Regression | Isolates benefit of recurrence vs. linearity vs. no model |
| Forward (out-of-time) hold-out    | Paper uses a static random-looking 80/20 split | Separate Jan–Jun 2026 evaluation set, never seen in training        | Tests true generalisation to the future                   |
| Diebold-Mariano significance test | Paper reports no statistical test              | Paired t-test on squared-error differentials vs. Naive baseline     | Adds statistical rigour beyond point estimates            |

## 16. Limitations

### 16.1 Limitations of the Original Paper

- Does not report which company, date range, or exact preprocessing/scaling scheme was used for the results in Section 11.1, harming reproducibility.
- Evaluates on a single static split with no repetition — no way to know if reported numbers are representative or lucky.
- Predicts raw/scaled price directly rather than returns, so headline accuracy is inflated by the autocorrelation of price levels (the same issue this report identifies in its own results, Section 13).
- No baseline comparison against a naive persistence model, so the reader cannot judge how much of the reported accuracy the LSTM actually adds.
- No statistical significance testing of the LSTM's advantage over Regression.

### 16.2 Limitations of Our Extension

- Single-stock study (TSLA only); conclusions about “which model is best” are specific to this stock's volatility profile and may not transfer to lower-volatility equities.
- Directional accuracy (49.5%) is close to chance overall — the model should not be interpreted as a trading-signal generator (Section 14).
- Macroeconomic features (Fed Funds Rate, 10-Year Yield, WTI, CPI, VIX, NASDAQ) are scaffolded in the code but not yet populated with real CEIC exports, so their potential contribution is untested (Section 17).
- As flagged in Section 12, the uploaded materials do not contain the executed Naive/Ridge numeric results, multi-seed standard deviations, or the Diebold-Mariano p-value, so the four-way comparison in this report is necessarily partial until that console output or CSV output is supplied.
- Reporting the “best seed” for the plotted figures (Figures 11.1–11.2) alongside the mean $\pm$ std for the tables is standard practice but should be read carefully: the plots are illustrative of a representative run, not of the central tendency across seeds.

## 17. Future Scope

- Populate the CEIC macroeconomic hooks already present in the code (Fed Funds Rate, 10Y Treasury Yield, WTI Crude, CPI) plus VIX and NASDAQ returns, and re-run the full pipeline to test whether macro-conditioning improves directional accuracy specifically in “choppy” months such as February and June 2026.
- Extend the comparative study to a small basket of stocks with different volatility profiles to test whether Model B v2's advantage over Model B v1 is Tesla-specific or general.
- Add SHAP or permutation-based feature-importance analysis to make the Bidirectional LSTM's predictions more interpretable to a non-technical audience.
- Explore attention mechanisms or Transformer-based sequence models as a further architectural comparison point against the Bidirectional LSTM.
- Complete the statistical reporting gaps flagged in Sections 12 and 16.2 (Naive/Ridge numeric results, seed-level standard deviations, Diebold-Mariano p-value) by re-running and archiving the full console/CSV output of `model_b_v2_improved.py`.

## 18. Conclusion

This project set out to replicate Parmar et al. (2018) and then go meaningfully beyond it. The replication is best characterised as conceptual rather than literal: both works apply LSTM networks to OHLCV-derived stock data, but nearly every implementation choice in Model B v2 — the log-return target, the 12-feature multivariate input, the Bidirectional architecture, the Huber loss, the forward-in-time hold-out, and the multi-seed statistical reporting — was deliberately changed to correct specific, identifiable weaknesses in the original paper's methodology. Applied to Tesla, the resulting model explains 80% of 2026 price variance while candidly reporting that its directional accuracy remains close to a coin flip, a result the report interprets through the lens of market efficiency rather than overselling as a trading edge. Model B v2 is recommended as the strongest available price-tracking tool for Tesla among the models evaluated, with the explicit caveat that several planned comparison numbers (Naive/Ridge results, seed-level variance, and the Diebold-Mariano test) still need to be supplied from an executed run of the underlying code before the analysis can be considered fully quantified.

## 19. References

- [1] I. Parmar, N. Agarwal, S. Saxena, R. Arora, S. Gupta, H. Dhiman, and L. Chouhan, “Stock Market Prediction Using Machine Learning,” 2018 First International Conference on Secure Cyber Computing and Communication (ICSCCC), Hamirpur, India, 2018, pp. 574–576.
- [2] M. Usmani, S. H. Adil, K. Raza, and S. S. A. Ali, “Stock market prediction using machine learning techniques,” ICCOINS 2016, Kuala Lumpur, pp. 322–327.
- [3] K. Raza, “Prediction of Stock Market performance by using machine learning techniques,” ICIEECT 2017, Karachi, pp. 1–1.
- [4] H. Gunduz, Z. Cataltepe, and Y. Yaslan, “Stock market direction prediction using deep neural networks,” SIU 2017, Antalya, pp. 1–4.
- [5] M. Billah, S. Waheed, and A. Hanifa, “Stock market prediction using an improved training algorithm of neural network,” ICECTE 2016, Rajshahi, pp. 1–4.
- [6] K. V. Sujatha and S. M. Sundaram, “Stock index prediction using regression and neural network models under non normal conditions,” INTERACT-2010, Chennai, pp. 59–63.

- [7] S. Liu, G. Liao, and Y. Ding, "Stock transaction prediction modelling and analysis based on LSTM," ICIEA 2018, Wuhan, pp. 2787–2790.
- [8] T. Gao, Y. Chai, and Y. Liu, "Applying long short term memory neural networks for predicting stock closing price," ICSESS 2017, Beijing, pp. 575–578.
- [9] K. A. Althelaya, E. M. El-Alfy, and S. Mohammed, "Evaluation of bidirectional LSTM for short- and long-term stock market prediction," ICICS 2018, Irbid, pp. 151–156.
- [10] Tesla Stock Price Prediction Economics Research Project — Methodology and Results (uploaded document, Tesla\_LSTM\_Economics\_Report.pdf), 2026.
- [11] model\_b\_v2\_improved.py — Model B v2 implementation source code (uploaded file), 2026.



## 20. Appendix: Data Gaps Requiring Author Input

Per the instructor's requirement to explicitly flag unavailable information rather than assume it, the following outputs are computed by `model_b_v2_improved.py` but were not present in the uploaded `Tesla_LSTM_Economics_Report.pdf`, and should be supplied (e.g. by re-running the script and sharing its console output or the `TABLE1_model_comparison.csv` / `TABLE3_DM_test.csv` files) to complete a fully quantified report:

- Exact numeric Naive-persistence RMSE / MAE% /  $R^2$  / Directional Accuracy for both the in-sample test split and the 2026 hold-out.
- Exact numeric Ridge Regression RMSE / MAE% /  $R^2$  / Directional Accuracy for both splits.
- Per-seed standard deviations for all Model B v2 metrics (currently only the aggregated/best-seed point estimates are available from the uploaded report).
- The Diebold-Mariano test statistic and p-value (TABLE3) comparing Model B v2 to the Naive baseline on the 2026 hold-out.
- Confirmation of which company's data underlies the original paper's reported  $R^2 = 0.86625$  and MSE figures, and the exact date range and scaler used — not stated in the paper itself.